

Docker

<https://docs.docker.com/desktop/setup/install/mac-install/>

<https://hub.docker.com/>

`docker --version`

`docker info`

nginx

-d detach mode, run in background

-p port mapping local port:8080, container port : 80

`docker run -d -p 8080:80 nginx`

Visit

<http://localhost:8080/>

Whalesay/Cowsay

`docker run docker cowsay Rohith`

Few Commands

`docker images`

`docker ps`

`docker ps -a`

Deleting a container

`docker rm <containerid>`

Alpine Container

-i interactive mode

-t stands for pseudo terminal

alpine is the image name

/bin/sh is the place where you want to run the image

```
docker run -it alpine /bin/sh
```

Inside the pseudo terminal, you can run all the linux commands

```
pwd
```

```
whoami
```

```
ls
```

```
exit
```

Provide Logs of a container

```
docker logs <container_id>
```

Remove Docker Image

```
docker rmi <image_id>
```

Note:

1. Always stop the container if it is running.

```
docker stop <container_id>
```

1. Then remove the container

```
docker rm <container_id>
```

1. After all these steps you could remove the docker image.

```
docker rmi <image_id>
```

Container Prune

Prune - Removing everything completely.

```
docker container prune
```

Pruning unused images

```
docker image prune -a
```

Pruning every images

`docker image prune`

Dockerizing an Application

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  Testing Simple Node App from Docker
</body>
</html>
```

server.js

```
const http = require('http');
const fs = require('fs');
const path = require('path');

const hostname = '0.0.0.0';
const port = 3000;

const server = http.createServer((req, res) => {
  const filePath = path.join(__dirname, 'index.html');
  fs.readFile(filePath, (err, content) => {
    if (err) {
      res.writeHead(500);
      res.end('Error loading file');
    } else {
      res.writeHead(200, {'Content-Type': 'text/html'});
      res.end(content);
    }
  });
});

server.listen(port, hostname, () => {
  console.log(`Server running at http://${hostname}:${port}/`);
});
```

Dockerfile

```
#My first docker build
```

```
#using debian as our base  
FROM debian:bullseye
```

```
WORKDIR /app
```

```
#installing node  
RUN apt-get update &&\  
    apt-get install -y curl &&\  
    curl -fsSL https://deb.nodesource.com/setup_22.x | bash - &&\  
    apt-get install -y nodejs
```

```
#Copying the required files  
COPY . .
```

```
EXPOSE 3000
```

```
CMD ["node" ,"server.js"]
```

```
# docker build -t simple-node-app .
```

Then run in cmd:

```
docker build -t simple-node-app .
```

Note:

t - tag name

. - search Dockerfile in current directory

The following command will make the image to run by creating container

```
docker run -d -p 3000:3000 simple-node-app
```

—name will be used to set the name of the container

```
docker run -d --name MyContainerName -p 3000:3000 simple-node-app
```

Then visit:

<http://localhost:3000/>

Note :

```
docker build -t nodeman-docker-example .
```

```
docker run -rm -it -v <physical path> nodemon-docker-example
```

rm: automatically remove containers before start executing

-it: for interactive pseudo output

-v: for volume mount